

Increasing and decreasing functions



1

a Decreasing

b Increasing

2

a 1

b $(1, \infty)$

c $(-\infty, 1)$

3 $(-\infty, 2)$

4

a

i $(-\infty, -2) \cup (2, \infty)$

ii $(-2, 2)$

b

i $(-1, 0) \cup (1, \infty)$

ii $(-\infty, -1) \cup (0, 1)$

5

a $(0, 0)$

b $(0, 0)$

c Increasing

6

a $(-1, 0)$

b Decrease

c Increase

Zeros of functions

7

a

i $(4, 0), (-2, 0)$

ii $4, -2$

b

i $(2, 0), (-4, 0)$

ii $2, -4$

c

i $(3, 0)$

ii 3

8

a -3

b $3, -1$

c $5, 3$



End behavior

9

a $(-4, 0), (0, 0)$

b $(0, 0)$

c ∞

d ∞

10

a The function does not have x -intercepts.

b $(0, 1)$

c ∞

11

a

i $(6, 0), (0, 0)$

ii $(0, 0)$

iii Increasing

iv Increasing

b

i $(-6, 0), (6, 0)$

ii $(0, 36)$

iii Decreasing

iv Decreasing

c

i $(-1, 0), (1, 0)$

ii $(0, 1)$

iii Decreasing

iv Increasing

d

i $(3, 0)$

ii $(0, 7)$

iii Decreasing

iv Increasing

e

i $(-4, 0), (0, 0), (4, 0)$

ii $(0, 0)$

iii Increasing

iv Increasing

f

i $(-2, 0), (-1, 0), (3, 0), (4, 0)$

ii $(0, 4)$

iii Increasing

iv Increasing

g

i $(-4, 0), (3, 0), (1, 0)$

ii $(0, -6)$

iii Decreasing

iv Increasing

h

i The function does not have x -intercepts.

ii $(0, 1)$

iii Increasing

iv Decreasing

i

i $(0, 0), (4, 0), (0, 3)$

ii $(0, 0)$

iii Decreasing

iv Decreasing

j

i (0,0)

ii (0,0)



iii Increasing

iv Decreasing

Asymptotes

12

a 0

b $y = 0$

c True

13

a 1

b $y = 1$

c True

14

a i $-\infty$ ii ∞ iii 0 iv 0

b $x = 0$

c $y = 0$

15

a i ∞ ii $-\infty$ iii 4 iv 4

b $x = 5$

c $y = 4$

16

a i ∞ ii ∞ iii 0 iv 0

b $x = 1$

c $y = 0$

17

a i 1 ii 1 b $y = 1$

c $(-\infty, \infty)$